

Programming Techniques

Week 1: Introduction to Programming



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| Today's Learning Journey



Concepts

Programming &
Algorithms Basics



Translation

Compilers vs
Interpreters



Python

Environment & Your
First Code



Debugging

Fixing Common
Mistakes

| What is Programming?

Instructions for Machines

Programming is the art of giving precise, step-by-step instructions to a computer to solve a problem.



Mobile Apps & Games



Banking Systems



Complex Data Analysis



| The "Making Tea" Algorithm

An Algorithm is simply a step-by-step recipe for solving a task.



Boil Water

Heat to 100°C



Add Tea

Leaves or bag



Sweeten

Add Sugar/Milk



Serve

Enjoy your drink

| Algorithm vs. Program

The Algorithm

A step-by-step logic/solution.
Language independent (English, Sinhala,
Flowcharts).



The Program

The algorithm written in a specific language (like
Python) so a computer can run it.



| How Computers Understand Us

Compiler

Translates the **entire** program first, then runs it.
(Like translating a whole book).



Interpreter

Translates and runs code **line-by-line**. (Like a live speech translator).

Python uses this!

| No Install? No Problem!



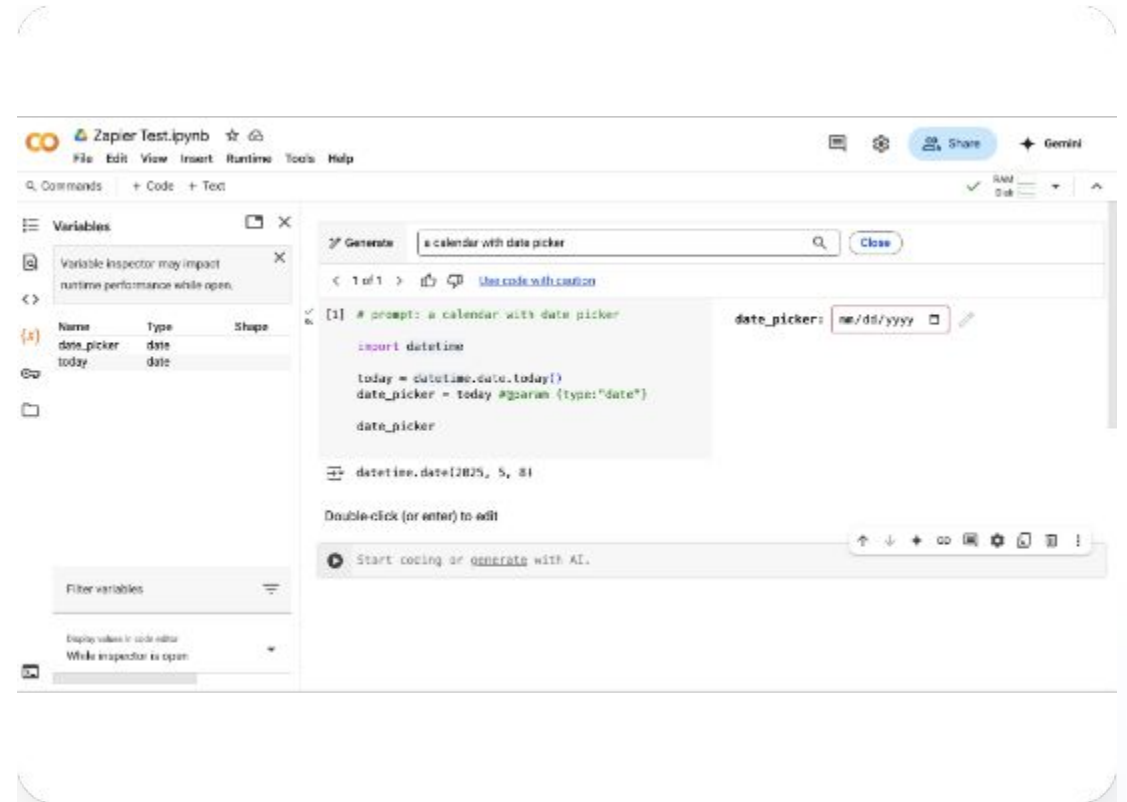
Google Colab: Use Google's servers for free.



Replit: Great for collaboration.



Benefits: Zero setup, works in browser, saves automatically.

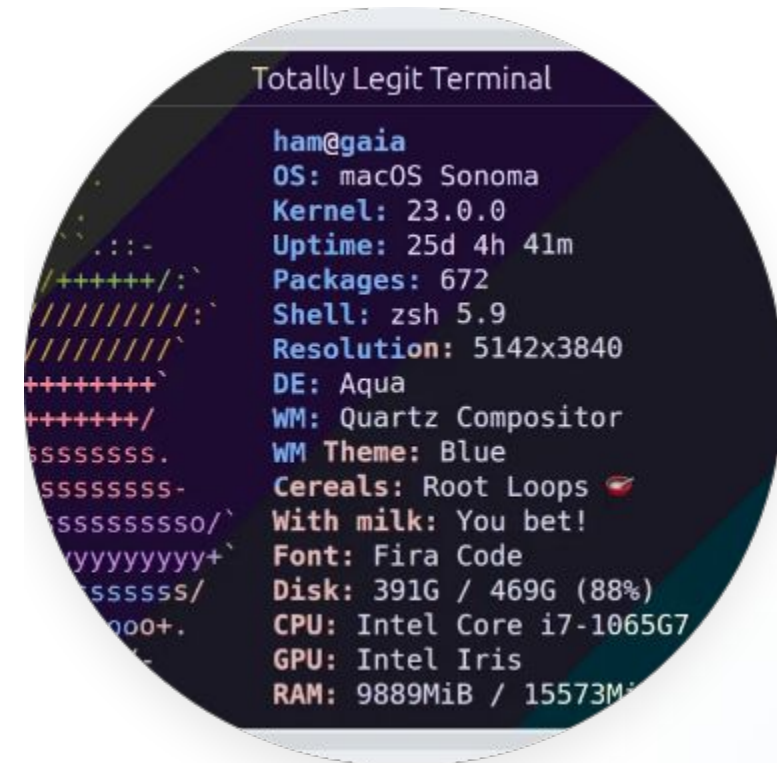


| The Hello World Program

```
print("Hello World")
```

What happened?

The `print()` function displays whatever is inside the quotes onto your screen.



| Writing Multiple Lines

```
print("Welcome to MIT") print("My  
name is Nimal") print("I am  
learning Python")
```

The Terminal Output:

```
Welcome to MIT  
My name is Nimal  
I am learning Python
```

| Quotes Matter!

```
print("5 + 3")
```

Result: 5 + 3

With quotes, Python treats it as literal **Text**.

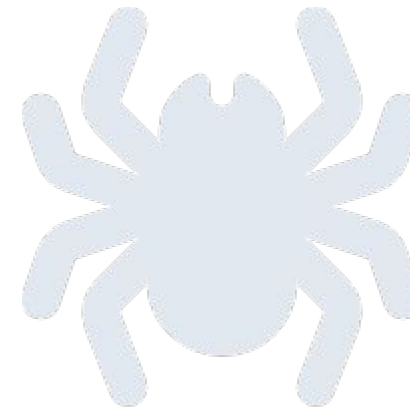
```
print(5 + 3)
```

Result: 8

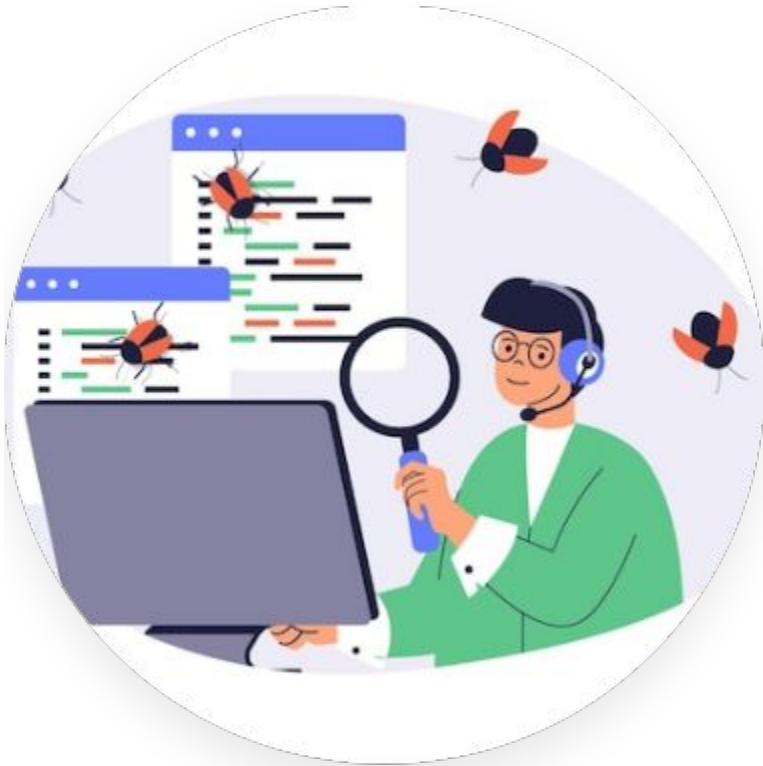
Without quotes, Python performs the **Calculation**.

| Watch Out for Bugs!

- “ Missing Quotes: Starting but not closing ".
Missing Brackets: Forgetting).
- T Case Sensitivity: Print is not the same as print.
- ⌨ Typing Mistakes: print instead of print.



| Can You Find the Error?



Spot the Mistake:

```
print("Hello World)
```

!Error: Missing the closing quote!

Python will stop and tell you exactly where it got confused.

| Practical Activities



Task 1

Print your full name and age on two lines.



Task 2

Print a personalized welcome message.



Task 3

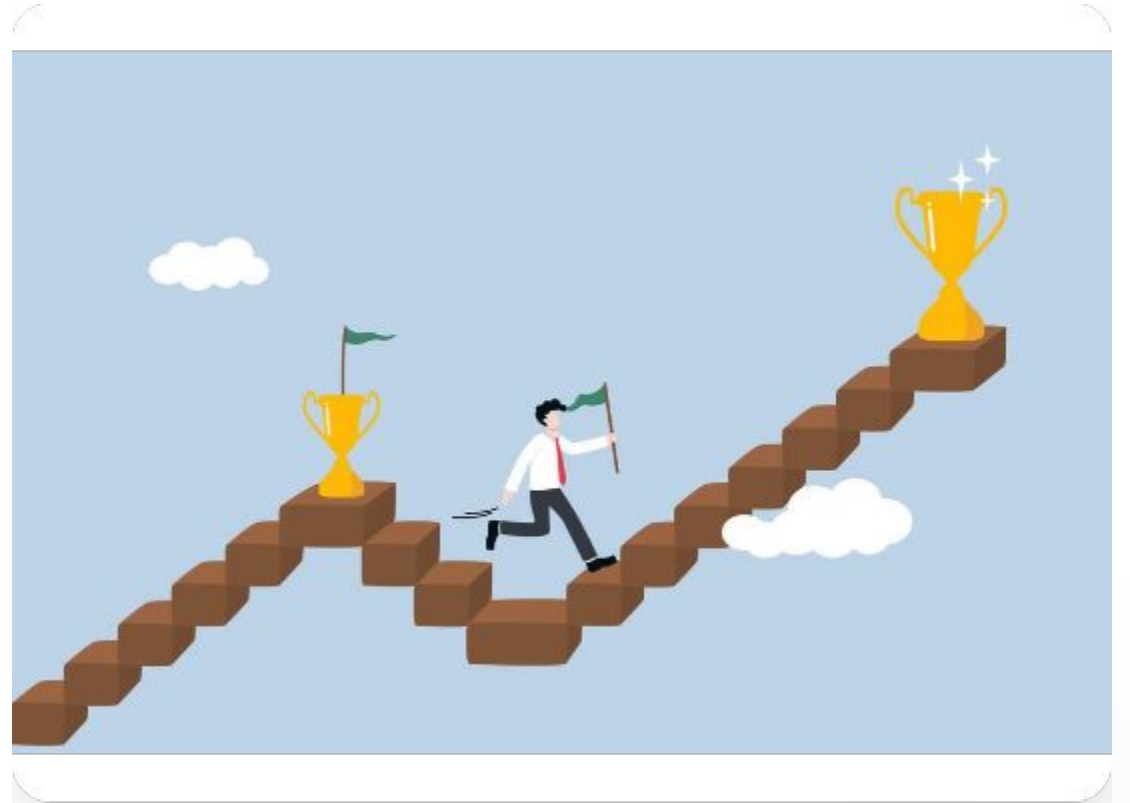
Create a pattern using '*' characters.

| The "MIT" Challenge

Goal: Print this exact header

```
-----  
WELCOME TO MIT PROGRAM  
Faculty of Graduate Studies  
Sabaragamuwa University  
-----
```

Hint: Use multiple print statements.



| Homework: Keeping it Practical



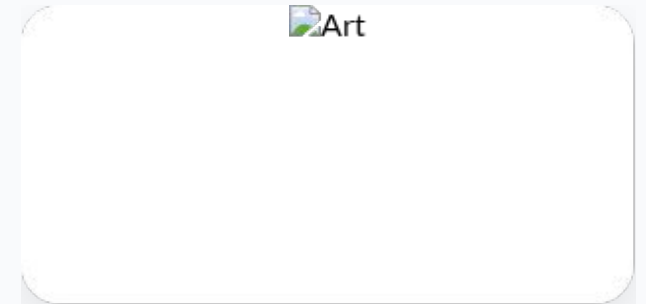
Schedule

Print your daily routine.



Favorites

Print a list of your top 5 movies.



ASCII Art

Draw a house or a star using
code.

| Image Sources



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Source: hamvocke.com



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| Image Sources



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